The Sandbox Project: The System Designer's Playground

Project Brief

Create a virtual playground where players can build creations for a predefined goal. The focus is on creating a set of "building blocks" and rules that players can combine for creative purposes. The game should provide the building system, and enough elements to create playful interactions. Combining the items should lead to surprising and unpredictable behavior. For players it should be an engaging experience that allows them to express themselves creatively and experiment with builds.

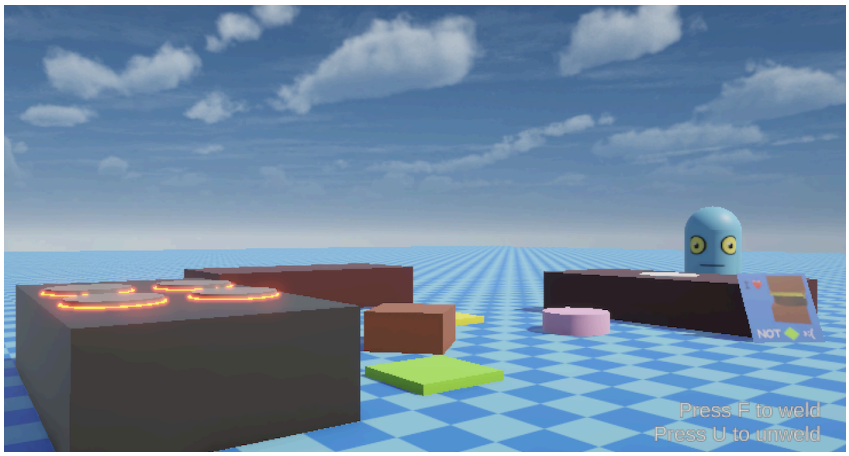
Decide which interactions between objects occur through specific rules and which interactions occur through a shared system rules (such as physics).

Learning Objectives

- Create emergent systems for building items.
- Motivate the player to experiment and build, and how to evoke a sense of discovery and fun.
- Apply playtest results to refine the player experience.

Deliverables

- A working Unity Build: A playable (webGL or multiplatform) version of your sandbox project on itch.io.
- Demonstration Video: A video (maximum 8 minutes, .mp4) in which you demonstrate the gameplay and the main systems of your game.
- The design report (see below)



Design report

The design document should contain the following items:

- How is the game played? What is the goal that players have to strive for?
- An overview of the items the user can build with, what are the properties of those items? What is the reason you added these building blocks?
- *A summary of the playtest results and a description of the iterations:* describe the key findings from the playtest. What feedback did you receive, and what observations did you make? Then explain what changes you made to the sandbox based on these findings and why.
- Screenshots of creative builds that players made.
- Please keep this max 6 pages.

Resources

- Sandbox Framework: https://github.com/acdh2/Sandbox

Requirements:

- Your project must contain at least three/four* object categories that can be combined with each other.
- For each category, a minimum of three/four* variations must be implemented.

**differentiation for solo/duo teams*

Your final grade for this project will be based on the provided rubric:

	Sufficient (required)	Good	Excellent
Sandbox Design	● The sandbox game provides mechanics and building blocks that allow players to experiment, create, and engage in playful interactions. ● The game has a clear objective and encourages players to explore their own ideas. ● Concrete examples of player-created content are shown.	One of the excellent criteria is met.	At least two of these criteria are met: ● A systematic/procedural approach was designed to efficiently create a large number of meaningful item variations. ● A clear and meaningful definition of the player experience was established and refined through multiple cycles of prototyping, testing, and improvement. ● The system allows building blocks to trigger chain reactions that result in unscripted outcomes.
Sandbox Implementation	● The student creates an implementation of the creative design. ● All core interactions are implemented and function without substantial problems.	One of the excellent criteria is met.	At least two of these criteria are met: ● Smooth, well-structured user experience from start to finish. ● Modular building block architecture allows the creation of variations efficiently. ● Unintended system behaviors, technical edge cases or emergent player actions are refined into stable mechanics.